

Travis Brown

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OBJECTIVE

Seeking an embedded/firmware engineering internship for Summer 2027.

EDUCATION

Rochester Institute of Technology

Bachelor of Science and Master of Science in Computer Science (Accelerated BS/MS Program)

GPA: 3.96

Rochester, NY

Expected: December 2027

SKILLS

Languages: C, C++, Python, Rust, Java, JavaScript, SQL, MIPS Assembly

Embedded: STM32 / ESP32 microcontrollers, FreeRTOS / ThreadX RTOS, CAN/CANopen, Modbus, MQTT, SPI, I2C, UART, PWM, ADC, OpenOCD, J-Link Segger, logic analyzer, oscilloscope

Tools: Git, GitHub, GitLab CI, Linux, macOS, KiCad, onShape, STM32 CubeMX / CubeIDE

WORK EXPERIENCE

Embedded Software Intern

GE Vernova

May 2026 – Present

- Developed embedded firmware in C for STM32F7 microcontrollers, implementing a full I/O subsystem from scratch for TransNEXT industrial radios with support for digital in/out, analog input, PWM output, and edge detection across three configurable I/O ports.
- Implemented Modbus RTU and ASCII server support from scratch, including a full register map exposing diagnostics, I/O states, and network health; designed and deployed an on-site customer-facing demo network with solar panels.
- Extended TransNEXT firmware with MQTT client support and optional TLS, including a CLI-based CA certificate upload and persistent storage flow for private broker authentication; enables remote diagnostics publishing and fleet-wide configuration updates over TCP.
- Debugged firmware and hardware integration issues using a Saleae logic analyzer, oscilloscope, and J-Link Segger.

Software QA Automation Intern

Lockheed Martin

May 2025 – August 2025

- Built a Python automation tool that reduced QA review time by 75% (20 mins → 5 mins) and automated 60%+ of manual validation checks; integrated into GitLab CI as an automated pre-review check phase on every PR.
- Automated convention scanning, commented-out code detection, dependency pinning via GitHub API, Jira ticket validation, and README parsing across multiple formatting styles.

PROJECTS

Autonomous Fixed-Wing Flight Controller

Personal Project

March 2026 – Present

- Programmed a custom flight controller from scratch on an ESP32S3 in C; RC passthrough, autonomous PID stabilization (roll and pitch), Madgwick AHRS sensor fusion for attitude estimation, and GPS-based heading navigation.
- Created a SITL (Software-In-The-Loop) simulator that models airframe dynamics, Gaussian sensor noise, and servo rate lag for iterative PID gain tuning.
- Designed custom PCB in KiCad; supports conventional and ailevon airframes via compile-time configuration.

Real-Time Vehicle Control Unit (VCU)

RIT Electric Vehicle Team

October 2025 – June 2026

- Engineered embedded C++ firmware for a custom STM32F446 PCB as the central VCU of an electric motorcycle; established CAN/CANopen communication with the motor controller, power subsystem, and battery management across dual CAN buses.
- Specified CANopen object dictionary and CAN message formats from scratch; built board-level fault handling, drive-by-wire control, and system health monitoring.
- Integrated a Simulink-generated 300 Hz state machine within ThreadX RTOS to meet hard real-time motor controller timing constraints.

Handheld Chip-8 Emulator Console

Personal Project

June 2025 – December 2025

- Wrote a Chip-8 emulator in Rust on a Raspberry Pi; designed and hand-soldered a custom 4x4 matrix keypad PCB in KiCad.
- Ported the emulator to Raspberry Pi hardware with a bare-metal SPI driver for the SSD1309 OLED; 3D-printed the enclosure in onShape.
- Composed a custom game-selection ROM in Octo assembly; implemented configurable cycle speed and per-ROM quirk flags to handle behavioral differences across Chip-8 variants.